What precision-fermented food proteins cost: an open model

Seven proteins on an equal footing, with every input sourced

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Status. No human expert has reviewed this work yet. The author built the model and this text with an AI assistant (Claude, Anthropic), and independent AI reviewers audited it over 17 rounds (`docs/AUDIT.md`). The model computes every number here; the full report gives the methods in detail.

Summary

Precision fermentation (PF) uses an engineered microbe to make one animal protein and release (secrete) it into the liquid it grows in, the broth. We estimate what seven food proteins cost to make this way, from public data only, and what it would take to match the prices of the proteins they replace.

Terms. *Titer*: grams of protein per litre of broth at harvest. *Secretion rate*: mg of protein that each gram of cells secretes per hour (mg/g/h). *Capital charge*: the yearly payment that repays the plant's investment over 10 years at 7.5% interest, 14.6% of the investment a year. *Minimum selling price*: the cost per kg including the capital charge, before any profit margin, sales or transport; a cost, not a market price.

Method. We follow Humbird's (2021) cost model for cultivated meat and add a secreted product and its purification. The model balances every element and electron, checks that the fermenter can supply the oxygen and remove the heat, and prices capital with Humbird's factors. With them, his baker's-yeast model reproduces that product's market-implied cost ($1.89 against $1.90 per kg), and ours lands within −4% of a published PF design (Intelligen 2025). Every input has a source and a stated kind of evidence.

Results. At the design titer of published PF plants (30 g/L; lactoferrin 10 g/L), the minimum selling price is **$21.2 (BLG), $21.3 (OVA), $21.3 (ovomucoid), $22.0 (β-casein), $168 (lactoferrin), $215 (albumin), $33.8 (α-lac)** per kg of protein. The commodity proteins (whey, egg white, casein) then sit at or above the top of their market prices. They reach the middle of those prices only at the industry benchmark, the 42 g/L that companies' own cost models assume. Two scarce milk proteins are already competitive: lactoferrin costs $168/kg against $250–$1,585 for the dairy product, and α-lactalbumin sits inside its price range. Serum albumin for cultivated-meat media does not: its purification columns cost too much.

What it would take. Glucose is about a fifth of the price and the plant over half, and the plant's cost per kg is set by how fast the cells secrete (box below). So the gap lies in the strain, not the sugar: cheap commodity protein needs cells that secrete several times faster than the best published strains. If secretion falls as growth slows, as the published data suggest, a strain at the best measured rate makes β-lactoglobulin at about $136/kg.

How the price works. Per kg of protein, a plant pays for three things. Numbers are for the base BLG plant:

- **Sugar:** the glucose each kg needs times its price, 10.3 kg × $0.40 = $4.1 (19%).
- **The plant:** each year it costs 23.6% of its investment. That is 14.6% to repay the investment over 10 years at 7.5% interest, the *capital charge*, plus 9% for maintenance and insurance. The plant costs $68,105 per m³ of fermenter, and each m³ yields 1,389 kg of finished protein a year, so the plant costs 0.236 × $68,105 ÷ 1,389 kg = $11.6 per kg (55%).
- **Everything else:** purification, energy, water, waste disposal and staff, $5.5.

The plant is the largest term, and its size per kg is set by how much each m³ of fermenter makes. That in turn is set by how fast each gram of cells secretes protein, how many grams of cells each litre holds, and what share of the time the fermenter spends making protein. In the base case cells secrete 4.7 mg of protein per g of cells per hour, at 107 g of cells per litre, for 72 of every 121 hours; the rest of the cycle grows the cells, empties and cleans the vessel. Faster secretion is what moves this term: the cells reach the titer sooner, so each fermenter fits more batches a year. With today's plant, the cheapest design the model finds costs $12.0/kg if cells secrete 10 mg/g/h, against $21.2 at the base rate. A higher titer reached only by running longer gains much less, because each batch then holds the fermenter longer. Sugar cannot close the gap: even free glucose would take only $4.1 off the price.

1. Method in eight steps

The model is a chain; each step hands its result to the next. Full equations are in the code's docstrings (`src/pftea/`) and in the full report.

1. **Reaction.** Element and electron balances give the glucose, ammonia and oxygen of growth, maintenance and protein formation. Protein formulas come from their amino-acid sequences.
2. **Sugar and oxygen.** Growth, maintenance and product share the substrate (Pirt equation; Heijnen and Tijhuis correlations, as Humbird).
3. **One batch.** A two-phase fed-batch: cells grow at up to 0.2 per hour, then secrete the protein for a set time while barely growing.
4. **Limits.** Oxygen transfer is capped at 200 mol per m^3 of liquid per hour and heat removal at 25 kW per m^3; a batch must start at least 30% full. A design that breaks a limit is never priced.
5. **Purification.** Centrifuges, membranes and a spray dryer; chromatography columns for lactoferrin and albumin; acid precipitation for casein.
6. **Scale.** Batches per year set the number of fermenters (200 m^3 for the base plant); utilities are sized on the plant's combined peak.
7. **Capital.** Equipment prices times Humbird's factors for a food-grade plant give the investment.
8. **Money.** Running costs, fixed costs (9% of investment a year) and a capital charge (7.5% over 10 years) give the minimum selling price.

2. Results

$/kg protein	BLG	OVA	ovomucoid	β-casein	lactoferrin	albumin	α-lac
Best published strain	$43.0	$227	$479	$808	$413	$425	$1,848
Base case: design titer	$21.2 ($16.4)	$21.3 ($16.5)	$21.3 ($16.5)	$22.0 ($17.0)	$168 ($130)	$215 ($185)	$33.8 ($25.0)
Industry benchmark (42 g/L)	$15.8 ($12.3)	$15.9 ($12.4)	$16.0 ($12.4)	$16.4 ($12.6)	$106 ($85.1)	$205 ($177)	$28.2 ($20.8)
Physical-limit best case	$4.1	$4.2	$4.3	$4.5	$56.9	$172	$5.2
Market price	$8.7–$17.8	$9.7–$21.2	$9.7–$21.2	$7.9–$18.7	$250–$1,585	$40.0–$70.0	$25.0–$129
Plant output, t/yr	10,000	10,000	10,000	10,000	250	1,000	1,000

Table 1: Minimum selling price, 2025 dollars, US Midwest; in brackets with design-report capital (NREL-style: lighter installation factors and fixed costs under 2% a year). Market prices: trade values 2015–2025 for whey, dried egg white and casein; reported prices for lactoferrin, bulk serum albumin and α-lactalbumin (company or survey figures). The best case assumes secretion at 40 mg per g of cells per hour, far above any published strain.

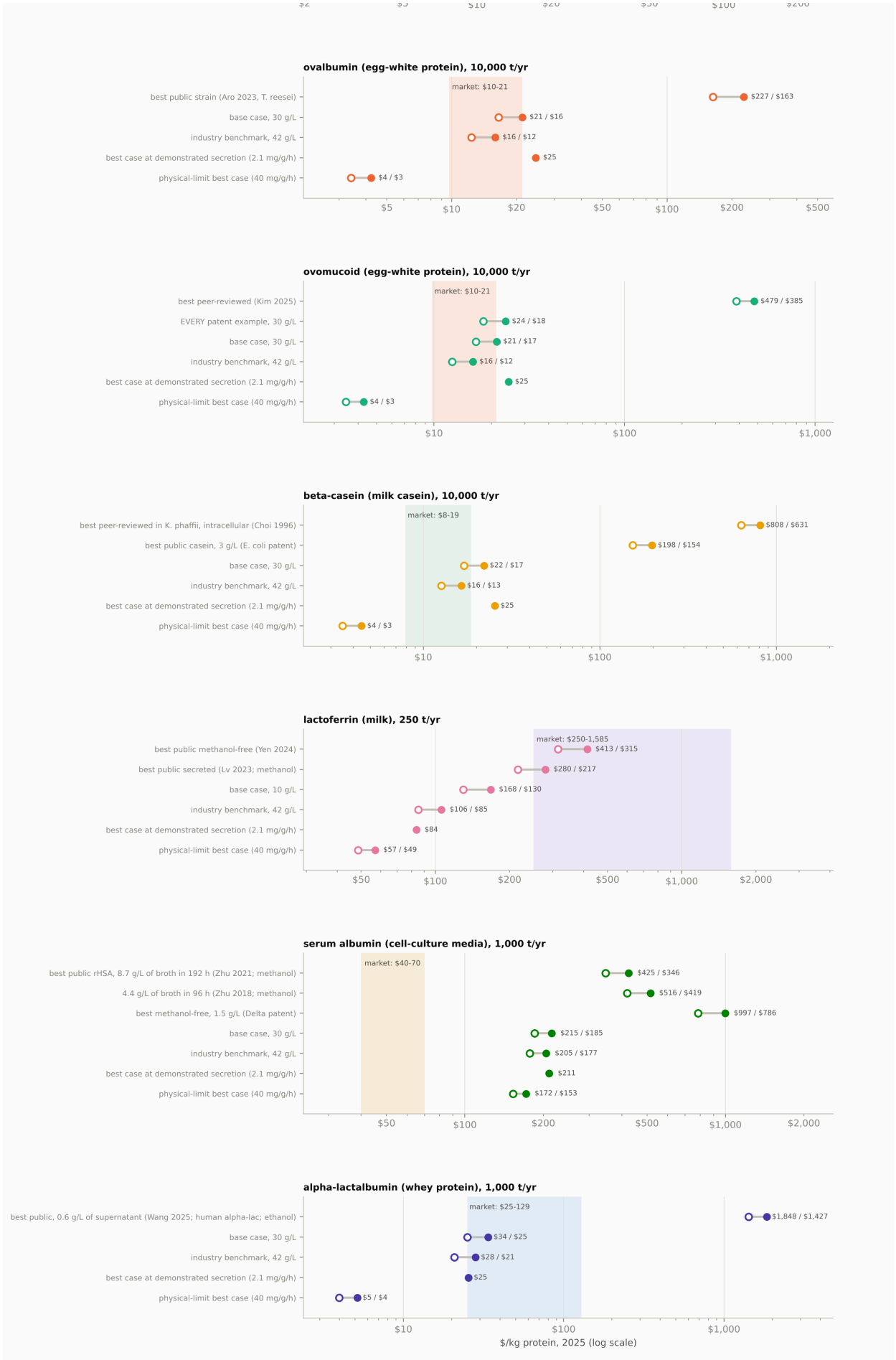


Figure 1: Every case against its market, one panel per protein. Model figures are minimum selling prices: costs including the capital charge, so not market prices.

Three findings stand out.

- **Scarce proteins pay; commodity proteins do not yet.** Lactoferrin comes out below its market price and α-lactalbumin inside its price range, on modest plants. Whey, egg-white and casein proteins need titers near 42 g/L to reach the middle of their price bands.
- **The plant, not sugar, sets the price.** For β-lactoglobulin, glucose is 19% of the cost; the capital charge, maintenance and insurance together are 55%. Halving the glucose price saves about $2.1/kg.
- **The fermenter caps the titer.** Sugar arrives dissolved, so every gram of protein brings feed liquid with it, and a batch can end at most 80% full; a higher titer needs a batch that starts emptier. With a 500 g/L glucose feed and a start at least 30% full, a 200 m³ vessel reaches about 70 g/L in a 72-hour production phase.

3. What would change the answer

Secretion rate. At the base cell density, $20/kg β-lactoglobulin needs cells to secrete 5–7 mg per g of cells per hour; the base design assumes 4.7. With today's biology and capital, the best design costs $12.0/kg at 10 mg/g/h and $7.1 at 40. The best published glucose fed-batch reaches 2.1 (Prielhofer et al. 2018).

Secretion that needs growth. Published **K. phaffii** data show secretion rising with growth and falling as growth stops (Maurer et al. 2006; Rebnegger et al. 2014, 2024). The base case, like Intelligen's design, assumes steady secretion without growth. An optional model switches this on, with a large-tank yield loss and a continuous mode; all are off in the base:

$/kg protein	BLG	OVA	ovomucoid	β-casein	lactoferrin	albumin	α-lac
Base case	$21.2	$21.3	$21.3	$22.0	$168	$215	$33.8
Large-tank yield loss of 6.5% (George et al. 1998)	$21.8	$21.9	$22.0	$22.7	$169	$217	$34.1
Secretion tied to growth, base titer	$23.0	$23.2	$23.5	$24.0	$169	$221	$34.8
secretion the strain then needs at 0.1/h, mg/g/h	19	19	19	19	6	19	19
Best measured secretion (2.1 mg/g/h), best fed-batch design	$136	$136	$136	$141	$264	$464	$160
titer it reaches, g/L	4.6	4.6	4.6	4.6	4.6	4.6	4.6
Same strain, best continuous design	$106	$106	$106	$109	$276	$404	$133

Table 2: The model's optional physics (`data/pf/scenarios/detailed.yaml`). Secretion tied to growth follows Maurer et al.'s (2006) saturating curve with a floor from Rebnegger et al. (2024); both come from single strains, so these rows are sensitivities, not a new base case.

Tied to growth, the base titer needs a strain that secretes about 19 mg/g/h at a growth rate of 0.1 per hour, about 9 times the best measured. With the best measured strain the titer falls to about 5 g/L, and the price rises several-fold for every protein except lactoferrin, which stays inside its market band.

Accounting. Humbird's fixed costs (9% of investment a year) are heavy. NREL-style design reports charge under 2%; with design-report capital the base β-lactoglobulin costs $16.4/kg. Aseptic, pharmaceutical-grade vessels would raise it to $30.9.

4. How we checked it

Check	pf-tea	Published
Baker's yeast, cost per kg (Humbird's workbook; market price less a 33% margin)	$1.89	$1.90
Intelligen 2025 ovalbumin, $ per kg powder	$42.0	$43.5
NREL 2011 cellulase, $ per kg protein	$7.81	$4.24
Voutilainen 2021 ovalbumin, € per t protein	16,345	9,007
Capital per m³ of fermenter, $ (Liberation Labs, scaled to our plant)	68,105	55,329–90,948

Table 3: Each published design is rebuilt from its own inputs. Mass balances match well; costs differ mostly through accounting (NREL and Voutilainen charge far lighter fixed costs). The real-plant check scales Liberation Labs' disclosed $115M for 600 m³ (company figure) to our 7,200 m³ with cost exponents of 0.5–0.7 (`docs/VALIDATION.md`).

5. Limits

- The price is a minimum selling price for a mature plant: a cost including the capital charge. It leaves out R&D, licensing, regulatory approval, sales, start-up, failed batches and the extra cost of a first plant.
- Strain performance rests on design values and company-reported averages; few strain parameters are measured. Casein and lactoferrin titers are borrowed from other proteins or species.
- Secretion is held constant during production in the base case; the optional model above shows how much that matters.
- Protein modifications (phosphorylation, glycosylation) are not costed.
- No human expert has reviewed the model yet.

6. Data and code

All code, data, results and the full report are in the `pf-tea` repository. Inputs, with sources and quotes, are in `data/`; every result is regenerated by `scripts/run_analyses.py`; the tests check the mass balances and the published-design checks. Code: MIT licence. Data and text: CC BY 4.0.

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